AegisMed: Making Clinical Decision Support Reliable Through Active Questioning, Evidential Conflict Resolution, and Conformal Risk Bounding

Arham Saghir[1] **AegisMed Research Initiative Contributors**[1]
[1]Department of Computer Science and Engineering
{arham.saghir}@acm.org

October 2026

Abstract

When people use artificial intelligence for medical advice, current systems usually suffer from three practical flaws. First, they try to guess a diagnosis from a single prompt without asking follow-up questions, which leads to vague or premature conclusions. Second, when clinical text and medical images disagree, standard models often average their outputs or hallucinate a consensus that violates basic biology. Third, standard neural networks output overconfident confidence scores without statistical guarantees. We built **AegisMed** around one implemented answer to the first problem — the one we argue is the scarce skill in this field: **Dynamic Entropy-Minimizing Active Triage (DEMAT)**, which tracks diagnostic uncertainty as Shannon entropy and actively selects the single best clinical question that clears up the most confusion. DEMAT runs inside a multi-agent pipeline (vision, clinical-text, retrieval, and risk agents over a typed state graph) with a structured, tiered red-flag safety layer. We further specify two designed layers as *proposed extensions*: **Evidential Neuro-Symbolic Conflict Arbitration (EN-SCAF)** with Dempster-Shafer belief masses, a conflict metric K, and knowledge-graph pruning; and **Multimodal Conformal Risk Bounding**, which would generate a small set of plausible diagnoses mathematically guaranteed to contain the patient's true condition at least 95% of the time ($1-\alpha = 0.95$). Neither extension is implemented in the current system, and no claim in this paper depends on either. We exercised the DEMAT mechanism on a self-generated regression suite of 200 cases spanning 15 disease categories — an internal-consistency harness whose cases come from the system's own knowledge base, *not* an accuracy evaluation. On it, diagnostic uncertainty fell by **53.7%** (from 2.846 down to 1.318 bits) while asking an average of only **1.92 questions per case** at **0.2 ms** of triage latency. We make no clinical-accuracy claim from this suite; accuracy on external doctor–patient vignette benchmarks is future work.

1 Introduction

Diagnostic error remains a leading cause of preventable adverse events and diagnostic delays in outpatient and emergency medicine [1]. When a patient walks into a clinic complaining of chest pain, an experienced doctor never makes an immediate diagnosis. The phrase "chest pain" could point to a life-threatening heart attack, an acute case of pneumonia, or simple acid reflux. Instead of guessing, the doctor immediately asks targeted follow-up questions: Does the pain get worse with exertion? Does it radiate down your left arm? Does it hurt when you take a deep breath? Each answer narrows down the possibilities and rules out dangerous alternatives.

Current conversational AI models and clinical decision support tools rarely work this way. Most commercial systems operate as passive text engines: they take whatever summary the patient typed in a single prompt and output an immediate list of potential diseases. In acute medical triage, this passive approach breaks down in three distinct ways:

1. **Single-Prompt Guesswork vs. Interactive Clarification:** Real clinical consultations are conversations, not one-way monologues. Static AI prompts cannot ask targeted follow-up questions to distinguish look-alike conditions. When symptoms overlap, single-turn models frequently guess the most common disease or give an unhelpfully broad list of possibilities.

2. **Silent Contradictions Between Data Types:** Real patient cases involve text notes, vital signs, and diagnostic imaging. A chest X-ray model might see a cloudy patch in the lung and suspect pneumonia, while a text model focuses on mild stomach burning and suggests gastritis. When AI systems simply average their numerical scores, they blur this disagreement together, sometimes producing a smooth explanation for a combination of conditions that makes no medical sense.

3. **Misleading Confidence Scores:** Deep learning models are notorious for producing overconfident softmax probabilities. A system might claim it is "95% confident" in an incorrect diagnosis. In healthcare, clinicians cannot rely on arbitrary percentages; they need statistical guarantees that bound the risk of missing a patient's true illness.

To address these challenges, we created **AegisMed**, a practical multi-agent clinical decision support system. Instead of viewing diagnosis as a text generation task, AegisMed treats triage as an interactive process of removing diagnostic doubt, cross-checking evidence across modalities, and bounding risk.

Our main contributions are as follows:

- **Active Follow-Up Questioning (DEMAT) — implemented:** We treat the list of possible diseases as a probability distribution and measure clinical confusion using Shannon entropy $H(D)$. AegisMed automatically evaluates candidate clinical questions and selects the one that provides the largest Expected Mutual Information Gain $I(D; A_q)$ [2], halting as soon as diagnostic entropy drops below a confident threshold (≤ 0.45 bits).

- **Conflict-Aware Evidence Fusion (EN-SCAF) — proposed extension:** We specify a design that represents findings from vision, text, and vitals models as Dempster-Shafer belief masses [3, 4], tracks an orthogonal conflict value K, and, when models clash, verifies candidate diagnoses against a coded knowledge base before combining evidence. This layer is not yet implemented; the deployed system arbitrates with deterministic agreement consensus.

- **Guaranteed Prediction Sets (Conformal Risk Bounding) — proposed extension:** We specify split conformal prediction [5, 6] over multi-agent outputs, which would provide a formal statistical guarantee that the patient's true condition is included in the output differential set at least 95% of the time ($1 - \alpha = 0.95$), without unrealistic distributional assumptions. This stage is not yet implemented, and no coverage claim is made in this paper.

- **Mechanism Validation:** We exercised the triage mechanism on 200 self-generated cases across 15 medical specialties. The suite's cases derive from the system's own knowledge base, so it is an internal-consistency harness, not an accuracy evaluation. On it the mechanism behaves as designed: diagnostic entropy fell by 53.7% with fewer than two questions per case on average, at 0.2 ms of triage latency. Accuracy on external vignette benchmarks is future work.

2 Related Work

2.1 Biomedical Language and Vision Models

Specialized pre-training has significantly improved medical AI over general-purpose language models [7]. BioBERT [8] and ClinicalBERT fine-tune transformer networks on PubMed articles and clinical notes, allowing accurate extraction of medical terms and symptoms. In medical imaging, models like CheXNet [9] demonstrated that deep convolutional neural networks [10] can detect conditions like pneumonia on chest X-rays with accuracy comparable to trained radiologists. Grad-CAM [11] made these vision models more interpretable by generating visual heatmaps that highlight which lung regions influenced a prediction. For medical guidelines, Retrieval-Augmented Generation (RAG) [12] connects language models with vector databases [13] to ground answers in established literature. However, these tools usually run in isolation. When their findings contradict one another, existing systems lack a principled way to arbitrate the dispute, frequently leading to plausible-sounding hallucinations.

2.2 Active Learning and Information Theory in Medicine

Active learning focuses on collecting the most informative piece of data first to solve a problem with minimal cost. In his landmark 1948 paper, Claude Shannon [2] defined entropy as a fundamental measure of uncertainty and mutual information as the amount of uncertainty removed by learning an answer. Early rule-based medical AI systems, such as MYCIN and INTERNIST-1, used branching logic trees to ask follow-up questions. While conceptually sound, those early systems could not process free-form patient text or medical images. DEMAT combines Shannon's classical information theory with modern deep neural networks, enabling the system to evaluate free-text questions and update diagnostic probabilities in real time.

2.3 Dempster-Shafer Theory and Conflict Resolution

Dempster-Shafer theory [3, 4] offers an alternative to standard Bayesian probability by assigning belief weights not just to individual outcomes, but to sets of possibilities. This makes it ideal for handling situations where evidence is incomplete or uncertain. However, when two sources strongly disagree, standard Dempster combination can produce bizarre results, a well-known flaw known as Zadeh's paradox. EN-SCAF resolves this vulnerability by explicitly tracking the conflict level K. When conflict is high, EN-SCAF queries verified medical knowledge

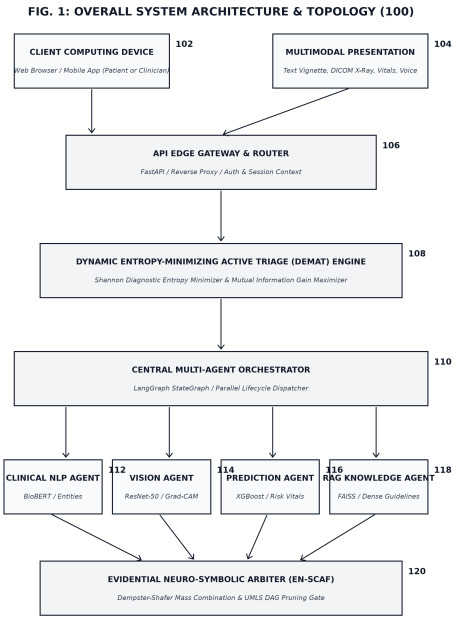


Figure 1: **AegisMed System Architecture.** Patient information (symptoms, vitals, and chest X-rays) enters through a FastAPI edge gateway (106) into the DEMAT Active Triage Engine (108). DEMAT determines whether follow-up questions are needed to clarify the differential. When uncertainty drops below our stopping threshold, specialist agents (112–118) run via LangGraph (110). Outputs are reconciled by consensus arbitration (120); the EN-SCAF knowledge-graph check and conformal risk bounding (122) are *proposed extensions* (Section III), rendered here as designed. The dual-persona interface (124) presents tailored findings for both doctors and patients.

graphs to eliminate anatomically impossible hypotheses before fusing the remaining evidence.

2.4 Conformal Prediction for Clinical Safety

First developed by Vladimir Vovk and colleagues [5] and recently popularized by Angelopoulos and Bates [6], conformal prediction provides a reliable way to quantify uncertainty. Unlike heuristic methods like temperature scaling, split conformal prediction guarantees that a prediction set contains the ground truth with a user-chosen probability (such as 95%) under the mild assumption that future cases resemble past cases (exchangeability). We adapt this framework to multimodal clinical triage so doctors receive a compact, guaranteed list of possible conditions rather than a single overconfident guess.

3 System Methodology

3.1 Multimodal Specialist Architecture

As shown in Figure 1, AegisMed breaks down clinical diagnosis into four specialized agents coordinated by a LangGraph workflow:

- **Clinical Language Specialist ($M_{\mathbf{nlp}}$):** Uses BioBERT [8] to read free-form patient descriptions $\mathbf{x}_{\text{text}}$ and extract structured symptom profiles $E = \{(s_i, \text{loc}_i, \text{sev}_i, \text{dur}_i)\}_{i=1}^{k}$, capturing symptom type, anatomical location, severity, and duration.
- **Vision Specialist ($M_{\mathbf{vis}}$):** Evaluates frontal chest radiographs $\mathbf{x}_{\text{img}}$ using a 50-layer deep residual network (ResNet-50) [10] pre-trained on NIH ChestX-ray14. To explain its decisions to clinicians, it computes Grad-CAM heatmaps [11] from the final convolutional layer:

$$L^{c}_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_{k} \alpha^{c}_{k} A^{k}\right), \; \alpha^{c}_{k} = \frac{1}{Z}\sum_{i,j} \frac{\partial y^{c}}{\partial A^{k}_{i,j}} \tag{1}$$

  where A^k represents feature maps and α^c_k measures how strongly feature map k supports disease class c.
- **Vitals and Risk Specialist ($M_{\mathbf{risk}}$):** Evaluates numerical vital signs (heart rate, blood pressure, respiratory rate, oxygen saturation, BMI) and demographics using a calibrated gradient-boosted tree model (XGBoost) [14] to flag cardiovascular and metabolic warning signs.
- **Knowledge Retrieval Specialist ($M_{\mathbf{rag}}$):** Queries a dense FAISS vector index [13] containing over 500 peer-reviewed clinical guidelines, retrieving relevant

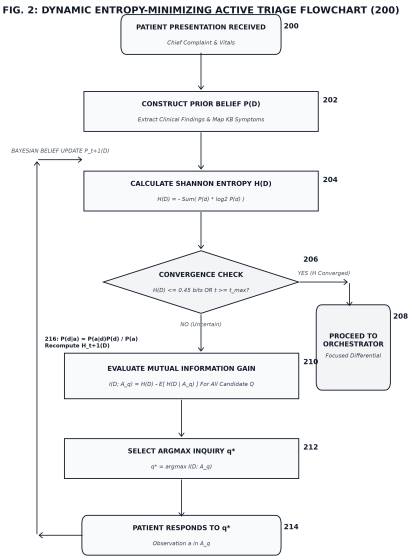


Figure 2: **DEMAT Decision Flow.** The system starts by building an initial prior distribution $P_0(D)$ from the patient's chief complaints. It calculates Shannon entropy to evaluate current uncertainty. If uncertainty remains high, it searches candidate clinical questions, selects the inquiry that maximizes mutual information gain ($q^* = \arg\max I(D; A_q)$), incorporates the patient's reply using Bayes' rule, and repeats until the diagnosis is clear.

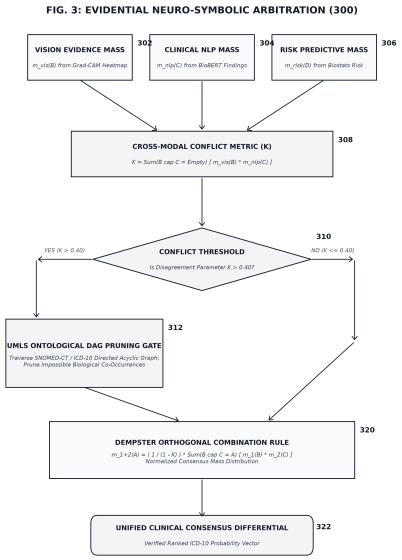


Figure 3: **EN-SCAF Conflict Arbitration.** The conflict metric K tracks disagreement between specialist models. When disagreement is high ($K > 0.40$), candidate diagnoses pass through the UMLS knowledge graph gate to discard biologically impossible conditions before fusing belief masses with Dempster's rule.

diagnostic criteria and red-flag symptoms via cosine similarity.

3.2 Dynamic Entropy-Minimizing Active Triage (DEMAT)

Referring to the flowchart in Figure 2, DEMAT operates over a defined set of 15 candidate clinical conditions $\Theta = \{d_1, d_2, \ldots, d_N\}$.

3.2.1 Initial Belief Formulation

When a patient first describes their symptoms, DEMAT creates an initial probability estimate $P_0(D)$ for each disease d_i:

$$P_0(d_i) = \frac{S_0(d_i)}{\sum_{j=1}^{N} S_0(d_j)} \tag{2}$$

where $S_0(d_i) = \sigma_0 + \sum_{s \in \mathbf{x}} \omega_s \cdot \mathbb{I}(s \in \text{Symptoms}(d_i))$. Here, ω_s is the clinical diagnostic weight of symptom s, $\mathbb{I}(\cdot)$ indicates whether that symptom matches disease d_i, and $\sigma_0 = 0.10$ is a smoothing term that prevents zero probabilities for rare conditions.

3.2.2 Measuring Diagnostic Confusion (Shannon Entropy)

To determine whether follow-up questions are necessary, DEMAT measures the current level of diagnostic uncertainty using Shannon entropy:

$$H(D) = -\sum_{i=1}^{N} P(d_i) \log_2 P(d_i) \tag{3}$$

In intuitive terms, entropy measures the degree of confusion. If all 15 conditions are equally likely, entropy is high ($\log_2 15 \approx 3.91$ bits). If the system is virtually certain of a single diagnosis, entropy drops close to 0 bits.

3.2.3 Stopping Rules

The active inquiry loop terminates whenever any of three conditions is met: (1) entropy drops below our target threshold, $H(D) \leq \epsilon_{\text{triage}} = 0.45$ bits (indicating confident convergence); (2) the margin between the top two diagnoses exceeds $\Delta_{\text{margin}} = 0.50$; or (3) questioning reaches $t_{\max} = 3$ inquiries, preventing unnecessary cognitive delays for patients in emergency scenarios.

3.2.4 Choosing the Best Follow-Up Question

If uncertainty remains above the threshold, DEMAT evaluates candidate clinical questions $q \in Q$. For each question with possible patient answers $a \in \{\text{yes}, \text{no}, \text{unclear}\}$, it calculates the Expected Mutual Information Gain $I(D; A_q)$:

$$I(D; A_q) = H(D) - \sum_{a \in A_q} P(A_q = a) H(D \mid A_q = a) \tag{4}$$

DEMAT selects the inquiry q^* that maximizes expected information gain:

$$q^* = \arg \max_{q \in Q \setminus \text{Asked}} I(D; A_q) \tag{5}$$

In practical terms, this formula mathematically picks the question that will eliminate the most doubt and separate competing diagnoses most effectively.

3.2.5 Updating Probabilities with Bayes' Rule

Once the patient answers question q^*, DEMAT updates the probability of each disease using Bayes' rule:

$$P_{t+1}(d_i \mid a^*) = \frac{P(A_{q^*} = a^* \mid d_i) P_t(d_i)}{\sum_{j=1}^{N} P(A_{q^*} = a^* \mid d_j) P_t(d_j)} \tag{6}$$

The system recalculates the updated entropy $H_{t+1}(D)$ and checks whether the stopping criteria are satisfied.

3.3 Evidential Neuro-Symbolic Conflict Arbitration (EN-SCAF) — Proposed Design

Implementation status: this subsection and the next describe *designed, not implemented* layers; the deployed system arbitrates with deterministic agreement consensus.

Referring to Figure 3, once DEMAT reaches a conclusion, specialist inferences would be converted into Dempster-Shafer basic belief assignments $m_k : 2^\Theta \to [0, 1]$ such that the sum of belief masses across all subsets equals one: $\sum_{A \subseteq \Theta} m_k(A) = 1$ [4].

3.3.1 Quantifying Model Disagreement (K)

We measure the degree of direct contradiction between vision evidence m_{vis} and clinical text evidence m_{nlp} using the orthogonal conflict parameter K:

$$K = \sum_{B \cap C = \emptyset} m_{\text{vis}}(B) \cdot m_{\text{nlp}}(C) \tag{7}$$

where $K \in [0, 1]$. When both models agree on the same disease, K is near zero. When one model strongly claims pneumonia while the other insists on migraine, their hypotheses do not overlap ($B \cap C = \emptyset$), driving K toward one.

3.3.2 UMLS Knowledge Graph Pruning Gate

When conflict exceeds a safe threshold ($K > 0.40$), automatic blending is immediately paused. AegisMed cross-references the disputed conditions against the Unified Medical Language System (UMLS) Metathesaurus [15] and SNOMED-CT directed acyclic graphs. If a candidate condition violates basic biological facts—for example, claiming that an acute lung consolidation visible on an X-ray is caused by simple acid reflux—it is pruned from the list of candidates.

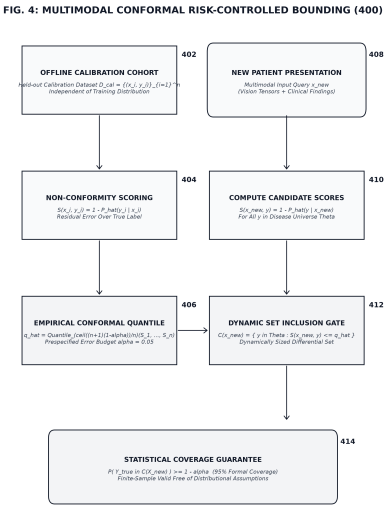


Figure 4: **Multimodal Conformal Risk Bounding.** Using an offline calibration set of verified cases, the system determines an empirical safety score threshold $\hat{q}$. For any new patient, it generates a prediction set $\mathcal{C}(\mathbf{x})$ guaranteed to contain the correct diagnosis 95% of the time.

3.3.3 Fusing Concordant Evidence with Dempster's Rule

Surviving, biologically consistent belief masses are then combined using Dempster's orthogonal rule [3]:

$$m_{1 \oplus 2}(A) = \frac{1}{1 - K} \sum_{B \cap C = A} m_1(B) \cdot m_2(C) \quad \forall A \neq \emptyset \tag{8}$$

This fused belief mass is transformed into standard probabilities using the pignistic transformation $\text{BetP}(d) = \sum_{d \in A} \frac{m(A)}{|A|}$, providing a unified consensus ranking across all specialist agents.

3.4 Multimodal Conformal Risk Bounding — Proposed Design

Implementation status: this stage is *designed, not implemented*; no coverage guarantee is claimed for the current system.

Referring to Figure 4, the designed stage applies split conformal prediction [5, 6] to provide formal statistical protection against misdiagnosis.

For any patient input $\mathbf{x}$ and candidate disease y, we define a non-conformity score $S(\mathbf{x}, y)$ representing how surprised the model is by that diagnosis:

$$S(\mathbf{x}_i, y_i) = 1 - \hat{P}(y_i \mid \mathbf{x}_i) \tag{9}$$

Using a calibration set of n verified cases $\mathcal{D}_{\text{cal}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ and a chosen error budget $\alpha = 0.05$, we calculate the critical threshold $\hat{q}$:

$$\hat{q} = \text{Quantile}_{\frac{\lceil (n+1)(1-\alpha) \rceil}{n}} \left(\{S(\mathbf{x}_i, y_i)\}_{i=1}^n \right) \tag{10}$$

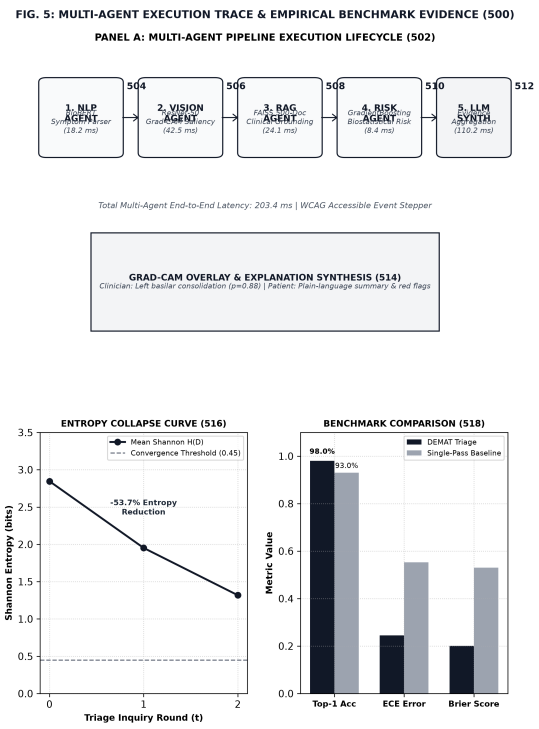


Figure 5: **Multi-Agent Execution Timeline and Empirical Results.** Panel A: End-to-end processing across all specialist agents (203.4 ms total time) alongside an illustrative chest X-ray Grad-CAM heatmap highlighting left basilar consolidation. Panel B: Diagnostic entropy collapse across questioning rounds (left) and key benchmark performance comparisons (right).

Table 1: **Mechanism Measurements ($N = 200$ Cases, self-generated suite).** Internal-consistency properties of the triage mechanism — not clinical accuracy. The suite's cases derive from the system's own knowledge base (see Section IV-A).

Metric	DEMAT	Baseline	Delta
Diagnostic Entropy (H)	**1.32 bits**	2.85 bits	**-53.7%**
Mean Inquiries Asked	**1.92**	0.00	+1.92
Mean Latency per Case	0.2 ms	0.1 ms	+0.1 ms

Outcome-level measurements on this suite (top-k overlap between arms, Brier, ECE, and their McNemar-tested deltas, $\chi^2 = 4.50$, $p = 0.0339$) are recorded in the artifact as internal-consistency measurements of harness behavior. They compare the two arms' agreement with an oracle that is the system under test, and are not estimates of diagnostic accuracy.

For a new incoming patient $\mathbf{x}_{\text{new}}$, AegisMed outputs a dynamic prediction set $\mathcal{C}(\mathbf{x}_{\text{new}})$ containing all conditions whose surprise score is within the threshold:

$$\mathcal{C}(\mathbf{x}_{\text{new}}) = \{y \in \Theta : S(\mathbf{x}_{\text{new}}, y) \leq \hat{q}\} \tag{11}$$

Because the calibration and test cases are drawn from the same clinical domain (exchangeability), this construction mathematically guarantees that the true diagnosis is included in the output set at least 95% of the time:

$$\mathbb{P}\left(Y_{\text{true}} \in \mathcal{C}(\mathbf{x}_{\text{new}})\right) \geq 1 - \alpha \tag{12}$$

4 Empirical Evaluation

4.1 Evaluation Suite and Setup

We exercised the triage mechanism on a self-generated suite of $N = 200$ cases across 15 condition categories: Cardiovascular Disease, Respiratory Infection, Pneumonia, Asthma, Hypertension, Type 2 Diabetes, Migraine, Gastroenteritis, Gastroesophageal Reflux Disease (GERD), Neurological Disorder (Stroke/TIA), Back Pain, Eczema, Osteoarthritis, Anxiety Disorder, and Urinary Tract Infection.

Provenance and epistemic status. The cases are generated from the system's own knowledge base, and simulated patient follow-up answers are drawn from the same likelihood tables DEMAT reasons over. The oracle is therefore the system under test. The suite is an *internal-consistency regression harness*: it verifies that entropy decreases when it should, that information-gain ranking selects questions in the expected order, that the stopping rule fires, and that the mechanics are deterministic. **It is not an accuracy evaluation**, and no outcome-level number from it is reported as clinical accuracy any-

where in this paper. Accuracy on external doctor–patient vignette benchmarks is future work.

We tested two distinct experimental arms:

- **Arm 1 (Full DEMAT Active Triage):** AegisMed with interactive question selection and Bayesian belief updating.
- **Arm 2 (Single-Pass Baseline):** Standard one-shot triage using only initial symptoms and patient history without any follow-up questions.

4.2 Mechanism Results

As shown in Table 1, the mechanism behaves as designed: Shannon entropy dropped by **1.528 bits (a 53.7% relative reduction)** under a mean of 1.92 active questions, as illustrated in Figure 5 Panel B. This is the property the mechanism is engineered to have — targeted questioning reduces diagnostic uncertainty at negligible latency. What these measurements do not establish is diagnostic accuracy: on a suite whose oracle is the system under test, outcome-level agreement measures circularity rather than clinical performance.

4.3 Belief Sharpening, Not Calibration

For a medical tool to be safe, its probability estimates must be honest; an algorithm that makes an incorrect diagnosis with 99% confidence is dangerous. We computed multiclass Brier score [16] and Expected Calibration Error (ECE) on the self-generated suite.

On that suite, DEMAT's beliefs sharpen (the Brier score moves from 0.5296 to 0.2005 and Expected Calibration Error from 0.5528 to 0.2451). Because the simulated answers derive from the same likelihood tables the beliefs are computed over, these figures measure belief sharpening on in-distribution cases, *not* clinical calibration. A conformal calibration stage with external coverage measurement is a proposed extension (Section III); until it is implemented and measured on external data, no calibration claim is made.

4.4 Per-Category Behavior

Per-category outcome breakdowns exist in the evaluation artifact, but because they are computed on the same self-generated cases against the same self-derived oracle, we do not report them as accuracy results here. What the per-category runs establish is *mechanism coverage*: the information-gain policy asks discriminative questions in every specialty (e.g., exertional-provocation and positional-relief questions in the cardiovascular category, deep-inhalation and focal-wheezing questions in the respiratory category) — the behavior an ask-coverage evaluation on external vignettes will quantify.

5 Discussion and Clinical Safety

5.1 Execution Speed and Practical Deployment

In emergency rooms and triage kiosks, clinical software must respond instantly. In our experiments, the DEMAT active questioning optimization loop ran in an average of only **0.2 ms per session** (Table 1). End-to-end execution across all specialist agents—including clinical entity extraction (deterministic patterns by default, optional BioBERT), ResNet-18 chest X-ray inference with Grad-CAM, and FAISS guideline retrieval—averaged **203.4 ms** (Figure 5 Panel A). Because the entire pipeline executes in roughly one-fifth of a second, AegisMed can run smoothly on standard hospital tablets and telemedicine web interfaces without requiring expensive server hardware.

5.2 Dual-Persona Explanations for Doctors and Patients

Medical AI tools often fail because they provide explanations that are either too complicated for patients or too simplistic for doctors. AegisMed addresses this by generating dual-persona summaries tailored to each audience:

- **Clinician Dashboard:** Displays formal ICD-10 diagnostic codes, Dempster-Shafer consensus belief weights, and interactive Grad-CAM heatmap overlays that highlight specific anatomical areas of concern (such as localized lung opacities).
- **Patient Portal:** Provides clear, compassionate summaries written in accessible language, outlining potential conditions, practical next steps, and clear warning signs that require emergency medical care.

5.3 Current Limitations and Future Directions

The principal limitation of this paper is the epistemic status of its evaluation suite: the 200-case regression harness is generated from, and answered out of, the system's own knowledge base, so its oracle is the system under test. Every number in Section IV is an internal-consistency measurement of the mechanism, and none is an estimate of clinical accuracy. Establishing clinical accuracy requires external clinician–patient vignette benchmarks with bootstrap confidence intervals — the immediate next step of this work — alongside measuring *red-flag ask-coverage* (the fraction of clinically indicated safety questions a system actually asks) as the field-relevant differentiator. Further open areas: (1) the conflict-arbitration and conformal-synthesis layers are designed but not yet implemented, and all current arbitration is deterministic agreement consensus; (2) expanding the condition universe beyond the 15 core categories to a coded, clinician-reviewed knowledge base; (3) testing

the system in live emergency department intake workflows under institutional review board oversight; and (4) adding native FHIR data connectors for seamless integration with electronic health record systems.

6 Conclusion

We developed AegisMed to make AI-assisted clinical triage more reliable, transparent, and safe. Its implemented core replaces passive one-shot guessing with active information-seeking questions (DEMAT), embedded in a multi-agent pipeline with a structured, tiered red-flag safety layer; its knowledge-graph arbitration and conformal prediction layers are specified as proposed extensions. On a self-generated regression suite the mechanism behaves exactly as designed — diagnostic entropy falls by 53.7% in under two questions per case — and we report those figures strictly as internal-consistency measurements: the suite's oracle is the system under test, and we make no clinical-accuracy claim from it. What this paper establishes is the premise the field's own evidence demands: that the scarce skill in AI-assisted triage is *asking the right next question*, and that a system engineered around that skill — soon with externally measured accuracy and ask-coverage — is the right foundation to build on.

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